Diabetes Digital Twin

Predicting glucose spikes 2 hours ahead by fusing EHR and CGM data

Team VVtech | PSG College of Technology, Coimbatore
Vinuba K (24PT37) - Team Leader
Vijay Adhithyan T (24PC39)

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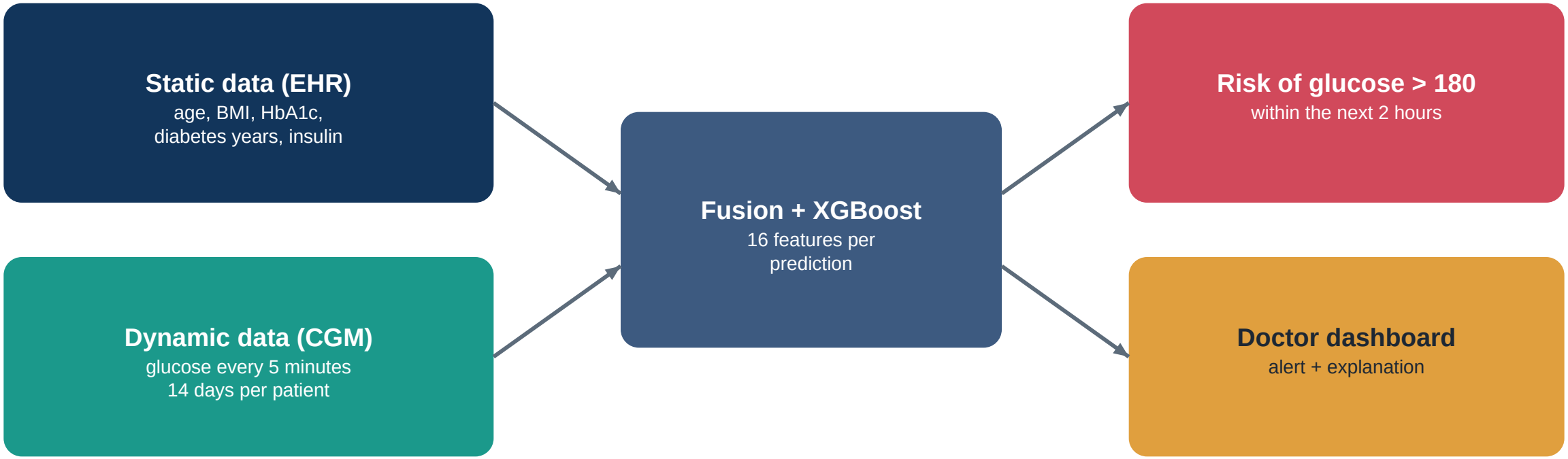
The Problem: Diabetes Care Is Reactive

- India has one of the world's largest Type 2 diabetes populations.
- Patients meet doctors only a few times a year; glucose spikes in between go unnoticed.
- Repeated spikes drive long-term complications: heart, kidney, eye and nerve damage.
- Need: shift from reacting to damage to predicting it before it happens.

Goal: warn the patient and doctor about a glucose spike 2 hours before it occurs

Our Solution: A Virtual Patient



Both data streams are fully synthetic, in line with DPDP Act and HIPAA sandbox rules.

Model and Method

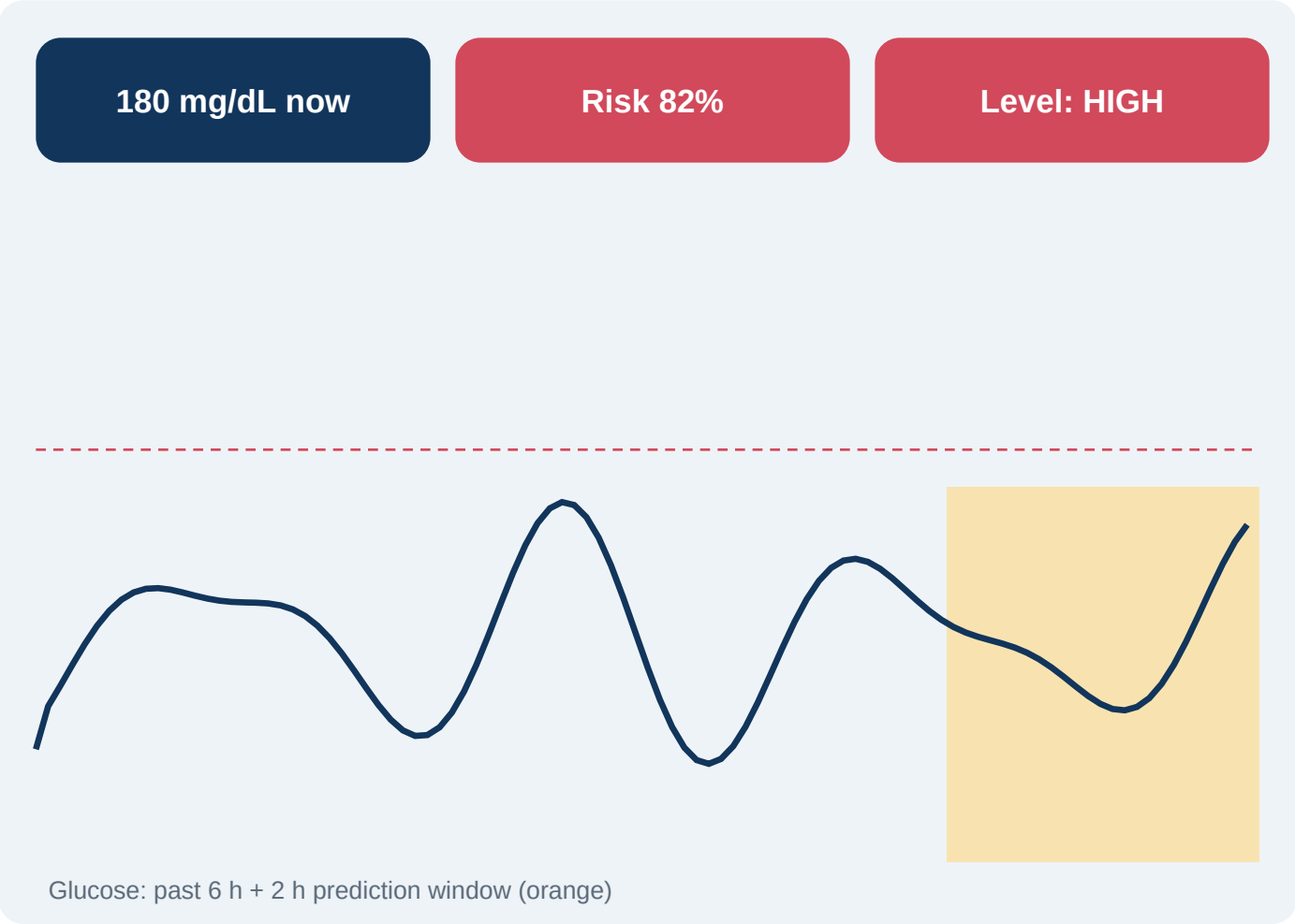
- Sensor features: current glucose; glucose 15 min, 30 min and 1 h ago; rate of change; 1-hour mean, std, max; time of day.
- EHR features: age, BMI, HbA1c, years with diabetes, insulin use.
- Target: will glucose exceed 180 mg/dL at any point in the next 2 hours?
- Model: XGBoost gradient-boosted trees, chosen for speed and explainability.
- Validation: split by PATIENT, so test patients were never seen in training.

Results (Synthetic Test Set)

0.985	0.935	0.880	0.881
AUC	Accuracy	Recall	Precision

- Most influential features: time of day, 30-min rate of change, current glucose, HbA1c.
- Honest note: synthetic data has regular meal patterns, so scores are optimistic. Real-world performance will be lower and needs validation on anonymized clinical data.

Doctor Dashboard (Streamlit)



- Pick a virtual patient and move through time.
- Live glucose trend with the 180 mg/dL limit.
- Risk score with LOW / MODERATE / HIGH alert.
- Patient EHR panel for context.
- 'Why this prediction?' chart shows which factors raise or lower risk.

Impact and Next Steps

- Impact: earlier warnings, fewer emergency visits, personalized lifestyle and insulin-timing advice, and better use of doctors' time.
- Next: add heart rate variability, sleep stages and step count; log meals and insulin.
- Validate on anonymized real-world data (MIMIC-IV, open CGM datasets).
- Personalize the model per patient over time; extend to hypertension and cardiovascular risk.
- License: MIT. Code: github.com/vinuba20/VVtech_PSG-College-of-Technology